Timur Javid

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Education

University of Illinois Urbana-Champaign (UIUC) **August 2022 – Present**

Doctor of Philosophy in Electrical and Computer Engineering

Advisors: Paul Kwiat and Eric Chitambar

University of Illinois Urbana-Champaign (UIUC) **August 2017 – May 2022**

Master of Computer Science (5-year BS/MCS) *GPA: 3.90*

Bachelor of Science in Computer Science

Relevant Coursework: Applied Machine Learning • Artificial Intelligence • Autonomous Vehicle Systems Eng. • Computer Vision • Distributed Systems • Information Theory • Quantum Cryptography • Quantum Info Processing Theory • Quantum Optics & Info • Quantum Optics & Devices • Quantum Opto-Electronics • Random Processes • Scientific Visualization

Technical Skills

Languages: Python • C • C++ **Tools & Skills:** SciPy • NumPy • PennyLane • Qiskit • PyTorch • Embedded Dev.

Presentations, Proceedings and Papers

- **T. Javid**, B. Doolittle, C. Lualdi, B. Karki, E. Chitambar, M. Vayninger, and P. Kwiat. *Variational Quantum Optimization with Optical Polarization Qubits.* Manuscript in preparation.
- L. M. Ramsey, K. Ortiz, S. J. Johnson, J. Krynski, R. Eason, N. Sultana, N. Bayat, Q. Lim, C. Jones, **T. Javid**, E. Widloski, M. Stefanini, J. Aller, B. Slezak, D. Suarez, S. Sachidananda, A. Ling, P. Battle, T. Jennewein, M. Lembeck, M. Mohageg, and P. G. Kwiat. *SEAQUE: UIUC-led Quantum Space Technology Demonstration.* Optica Quantum 2.0 Conference, June 2025. DOI: 10.1364/QUANTUM.2025.QTu2A.1
- A. Conrad, R. Cochran, D. Sanchez-Rosales, S. Isaac, **T. Javid**, T. Rezaei, A.J. Schroeder, G. Golba, A. Gutha, B. Wilens, K. Herndon, A. Hill, J. Chapman, I. Call, J. Szabo, A. Corrigan, L. Kamin, N. Lütkenhaus, D. J. Gauthier, and P. G. Kwiat *Drone- and Vehicle-Based Quantum Key Distribution* Manuscript under review at Nature Communications. arXiv preprint arXiv:2505.17587 (2025).

Professional Experience

Aliro Quantum **May 2025 – Dec 2025**

Simulation Team Intern, Aliro Quantum Network Simulator

- Implemented quantum process tomography using Aliro's quantum network simulator to serve as an example application for customers.
- Integrated the quantum network simulator with Pennylane, allowing for the use of Pennylane's quantum circuit and optimization framework for network optimization tasks.

IBM Research **May 2023 – August 2023**

Quantum Researcher Intern, Quantum State Transfer Systems

- Produced an experimental quantum optical set-up for ancilla-assisted quantum process tomography with spontaneous parametric downconversion (SPDC) waveguides and superconducting nanowire single photon detectors (SNSPD) in Bluefors dilution refrigerator.
- Cooperated with experimentalists and theorists on analysis of novel quantum state transfer systems.
- Analyzed novel quantum state transfer systems by implementing and executing quantum process tomography protocols, using Qiskit to gather data.
- Hands-on experience with fiber splicing, opto-mechanics, dilution refrigerator bring-up, optical alignment, and set-up and control of SPDC waveguides and SNSPDs.

IBM Research **May 2022 – August 2022**

Quantum Researcher Intern, Quantum Observability

- Incorporated three live data sources from quantum systems monitoring into a backend data pipeline utilizing Apache, Kafka, MQTT, and InfluxDB in Python, enabling observability for IBM Quantum systems.
- Enhanced data logging for existing alerts and notification system with a new data pipeline, bypassing logging to hard disk.
- Analyzed historical data to develop data visualizations with feedback from quantum researchers across teams.
- Constructed a front-end interface with the Carbon/React framework to display live data from quantum systems, linked with our InfluxDB backend.

Amazon Web Services (AWS) — May 2020 – August 2020

Software Development Engineer Intern, IoT Edge Qualification

- Spearheaded and developed a front-end user interface for an existing hardware qualification tool using React and Redux and a Golang backend within two months.
- Developed a web API in Go, allowing device tests to be run from HTTP requests and allowing users to obtain results from qualification tests.
- Integrated web API with a React/Redux front-end, providing users a visual representation of device tests, documentation for qualification standards, and feedback for which IoT services their hardware is compatible with.
- Led design meetings for the project, producing design documents and integrating feedback from code reviews and weekly meetings with peers and supervisors.

Academic Experience

Kwiat Quantum Information Group at UIUC — May 2021 – Present

Graduate Research Assistant

- Integrated OpenCV detection and multi-threaded C++ code with PID control to improve drone-to-drone locking for quantum key distribution, decreasing average quantum channel loss during flight from 20.6 dB to 9.68 dB.
- Designed and implemented firmware on an STM32 micro-controller for entanglement source experiment to be tested on the International Space Station, in collaboration with the Laboratory for Advanced Space Systems at Illinois.
- Analyzed performance of variational quantum optimization of polarization qubits for mitigating fiber noise, setting up a quantum optical experiment and automating control of the experiment.
- Maintaining an open-source Python package (Quantum-Tomography) and webpage utilizing SciPy optimization for quantum state and process tomography of optical systems, deployed as a public webpage for use by experimentalists.

Course Staff - ECE 498EC Quantum Information Processing — August 2024 – December 2024

Teaching Assistant

- Collaborated with professor to improve introductory course content to enhance student understanding.
- Tutored students one-on-one during office hours to cement understanding of course concepts and guided them through problem solutions.
- Organized course materials, graded problem sets and exams, and provided extensive and timely feedback to students using Canvas.

ScribeAR Group at UIUC — September 2019 – August 2022

Student Researcher

- Maintained a codebase for an augmented reality (AR) compatible web application providing live captioning services, utilizing Web Speech API and integrating Microsoft Azure captioning in a React and Redux front end.
- Prototyped hardware using Arduino for sound localization to provide users with directionality in a heads-up display.
- Formulated an experiment to gather feedback on effectiveness of AR-based captioning in academic settings.

Extracurriculars

Midwestern Robotics Design Competition — August 2019 – May 2021

Director

- Built and maintained the organizational website to provide teams and sponsors with easy access to information.
- Established relationships with other corporations for fundraising and sponsorships and worked closely with the Engineering Open House committee for event logistics.
- Organized community outreach events, presenting at a local children's science museum to get children interested in robotics.
- Initiated media coverage of the competition to increase promotion and awareness.

Miscellanea

Hobbies & Interests

- Hiking & Backpacking: Backpacking and hiking is a great way to unwind and disconnect. I frequently look for new hiking trails locally and drive out to new places.
- Photography: You can check out some of my photos on my website! Taking photos in nature helps me appreciate my surroundings.
- Cooking: I see cooking as a way to connect with other people and cultures. I like to experiment with different cuisines and cooking styles. One of my favorite foods to make (and eat) is Neapolitan style pizza.